

Dental Caries & Current Concepts of Etiology

Part 1: Dental Caries – Overview, Classification & Theories

1. Definition of Dental Caries

- **WHO Definition:** A localized post-eruptive pathologic process of external origin involving softening of the hard tissue and proceeding to the formation of a cavity.
- An irreversible, infectious, microbiologic disease of the calcified tissues of teeth, leading to demineralization of the inorganic portion and destruction of the organic substance.

2. Classifications of Dental Caries

A. Based on Rate of Progression

- **Acute Caries:** Rapid progression, lighter in color, often leads to early pulp exposure.
- **Chronic Caries:** Long-standing/slow progression, dark brown/black, usually asymptomatic.
- **Arrested Caries:** Static lesion with brown pigmentation, characterized by sclerotic/tertiary dentin formation.

B. Based on Anatomical Site

1. **Pit and Fissure Caries (Occlusal):** Occurs in deep narrow grooves.
2. **Smooth Surface Caries:** Occurs on proximal surfaces or cervical areas.
3. **Linear Enamel Caries:** Atypical destruction on labial surfaces of anterior teeth associated with metabolic defects (e.g., hypocalcemia) or birth trauma.
4. **Root Caries:** Occurs on exposed root surfaces; progresses rapidly due to absence of protective enamel layer, closer to pulp, and harder to restore.

C. Based on Occurrence / Sequence

- **Primary (Initial) Caries:** Initial lesion on a previously intact tooth surface.
- **Secondary (Recurrent) Caries:** Occurs adjacent to or beneath an existing restoration.

D. Based on Extent / Depth

- **Incipient Caries ("White Spot Lesion"):** Subsurface demineralization with an intact outer surface layer. Non-cavitated, reversible via remineralization (fluoride & sealants).
- **Hidden (Occult) Caries ("Fluoride Bombs"):** Occlusal dentin caries missed visually due to remineralized/intact enamel surface, but detectable radiographically.
- **Cavitation:** Breakdown of the enamel surface forming an open cavity.

E. Based on Tissue Involvement

1. **Initial Caries:** Demineralization without structural loss (reversible).
2. **Superficial Caries (*Caries superficialis*):** Confined to enamel.
3. **Moderate Caries (*Caries media*):** Extends into dentin.
4. **Deep Caries (*Caries profunda*):** Deep dentin lesion close to the pulp.
5. **Deep Complicated Caries:** Caries resulting in pulp exposure.

F. Based on Pathway of Spread

- **Forward Caries:** Spreads from enamel to dentin.
- **Backward Caries:** Caries spreading laterally in dentin, undermining and attacking enamel from the dentinal side.

G. Based on Chronology / Age Group

- **Early Childhood Caries (ECC):**
 - *Nursing Bottle Caries:* Associated with improper bottle feeding/pacifiers with honey; **spares mandibular incisors.**
 - *Rampant Caries:* Rapidly spreading decay affecting primary/permanent teeth, **including mandibular incisors.**
- **Adolescent Caries:** Rapidly burrowing type with large pulp chambers causing early pulp involvement.
- **Adult Caries:** Typically seen in older adults (55–60 years) with gingival recession and reduced salivary flow, manifesting as root/cervical caries.

Comparison Between Nursing Caries & Rampant Caries

Feature	Nursing Caries	Rampant Caries
Age Group	Seen in infants and toddlers.	Seen in all ages, including adolescence.
Dentition Affected	Affects primary dentition.	Affects both primary and permanent dentitions.
Involvement of Mandibular Incisors	Mandibular incisors are NOT involved.	Mandibular incisors ARE affected.
Etiology (Causes)	• Improper bottle feeding. • Pacifiers dipped in honey or other sweeteners.	• Multifactorial. • Frequent snacking. • High consumption of sticky refined carbohydrates. • Decreased salivary flow rate. • Genetic background / predisposition.

H. Based on Number of Surfaces Involved

- **Simple:** Involves 1 surface.
- **Compound:** Involves 2 surfaces.
- **Complex:** Involves >2 surfaces (e.g., MOD).

I. WHO System Scoring (4-Point Scale)

- **D1:** Clinically detectable enamel lesions with intact surfaces.
- **D2:** Cavities limited to enamel.
- **D3:** Cavities extending into dentin.
- **D4:** Lesions extending into the pulp.

3. Special Types of Caries

- **Radiation Caries:** Rampant caries following radiotherapy to the head and neck due to severe xerostomia (dry mouth), reduced pH, and increased saliva viscosity.

4. Theories of Dental Caries

Historical theories include:

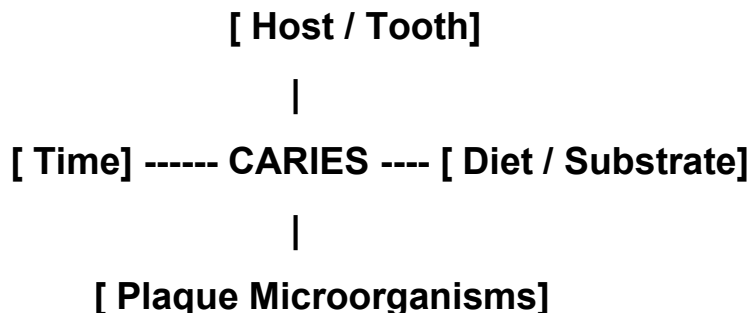
- Worm Theory.
 - Vital Theory.
 - Chemical Theory.
 - Parasitic Theory.
 - Proteolytic Theory.
 - Proteolytic-Chelation Theory.
- **Acidogenic / Chemico-Parasitic Theory (W.D. Miller, 1890)**
 - Most widely accepted classic theory.
 - Two-stage process:
 1. Decalcification of enamel/inorganic portion by bacterial acid attack.
 2. Dissolution of the remaining organic soft matrix by proteolytic bacterial action.

Part 2: Current Concepts of Caries Etiology

The Four Primary Interrelated Factors

For a carious lesion to develop, all **4 primary factors** must act simultaneously:

Tooth (Host)+ Bacteria (Biofilm)+ Diet (Substrate)+ Time= Carious Lesion



Factor 1: Diet & Substrate

- **Frequency vs. Quantity: Frequency of sugar intake** is more critical than total quantity. Constant snacking causes repeated acid drops below critical pH.

- **Physical Consistency:** Sticky carbohydrates adhere to tooth surfaces longer, maintaining prolonged acid exposure.
- **Xylitol:** Non-cariogenic & cariostatic sweetener. Reduces plaque volume, lowers acid output, and inhibits *Mutans Streptococci* growth.

Factor 2: Host Resistance

1. Tooth Factors:

- Deep pits and narrow fissures trap bacteria and food debris easily.
- Tooth malalignment hinders self-cleansing and plaque removal.

2. Salivary Protection:

- **Buffering Capacity:** Bicarbonate ions neutralize bacterial acids.
- **Flow Rate (~750 mL/day):** Helps clear sugars and debris.
- Decreased flow (Xerostomia) significantly increases caries susceptibility.

Factor 3: Plaque Microorganisms

- **Mutans Streptococci (MS): Primary Initiator** of dental caries. Produces extracellular glucans for firm tooth attachment and rapidly ferments sugars to lactic acid.
- **Lactobacilli (LB): Secondary Continuer.** Highly aciduric (survives in low pH environments); drives the progression of deep cavitated lesions.

Factor 4: Time Requirement

- Caries is a slow process requiring prolonged, uninterrupted exposure to acid on susceptible surfaces.